Step-by-Step Commands to Run the Project

Team Members: 1. Kushal G (PES1UG24AM145) | 2. Kshitij Satish Shetty (PES1UG24AM143)
Open PowerShell or Command Prompt (CMD) inside the **BigData_Fraud_Detection** folder and run these commands in order.

PART A: Main Commands to Run on Any Laptop (No Docker Needed)

Command 1: Run Automated Tests (Checks Code & Model Accuracy)

```
python -m pytest -v
```

What it does: Checks that transaction JSON data is valid, proves the 'Class' fraud answer is never leaked into model inputs, checks that Flink gives the exact same fraud probabilities as scikit-learn, and checks Apache Iceberg storage.
What you will see: `14 passed` in green.

Command 2: Run the 3-Speed Streaming Test into Apache Iceberg

```
python scripts/smoke_test.py
```

What it does: Streams 600 test transactions at 3 increasing speeds (**200, 500, and 1,000 transactions/sec**) through the Flink scoring function into a local **Apache Iceberg** table. Also injects 5 corrupted messages per run to prove our system isolates bad data without crashing.
What you will see: `[OK] Smoke test PASSED across 3 rates: 600 total valid events persisted to Apache Iceberg...`

Command 3: Print Fraud Model Accuracy & 3-Segment Table

```
python scripts/run_evaluation.py
```

What it does: Evaluates all **56,962 held-out transactions** (containing 98 actual frauds) and prints the Confusion Matrix (TP=90, FP=1386, TN=55478, FN=8), Recall (91.84%), False-Positive Rate (2.44%), and the **Low (≤ $9.99)**, **Medium ($9.99–$50.00)**, and **High (> $50.00)** segment error table.

Command 4: Verify the 4-Page Technical Report PDF

```
python scripts/generate_report.py
```

What it does: Verifies (or rebuilds) the 4-page `report/technical_report.pdf` report.

Command 5: Launch the Simple Presentation Web UI (Frontend)

```
python scripts/presentation_ui.py
```

What it does: Starts the simple web dashboard at **http://localhost:8501** and opens it automatically in your browser so you can show live transaction fraud checking, segment tables, and load-test runs.
(Press Ctrl + C in the terminal when you want to stop the UI server.)

PART B: Optional Commands (Download Dataset & Train From Scratch)

```
# 1. Download the 150 MB creditcard.csv dataset from Zenodo
python scripts/download_dataset.py

# 2. Train Logistic Regression model on 80% data and test on 20% data
python scripts/train_model.py --threshold 0.50
```

PART C: Full Docker Cluster Commands (If Docker Desktop Is Running)

```
# 1. Start Kafka, Flink, Iceberg REST Catalog, and MinIO containers
docker compose up -d --build
docker compose ps

# 2. Create the Kafka topic for transactions
docker exec fraud-kafka /opt/kafka/bin/kafka-topics.sh \
  --bootstrap-server kafka:9092 --create --if-not-exists \
  --topic transactions.raw.v1 --partitions 2 --replication-factor 1

# 3. Submit the Flink real-time fraud scoring job inside the Flink cluster
docker exec -d fraud-flink-jobmanager ./bin/flink run \
  -py /opt/app/src/flink_job.py --bootstrap-servers kafka:9092 \
  --iceberg-rest-uri http://iceberg-rest:8181 --s3-endpoint http://minio:9000

# 4. Run the 3-speed load test (200, 500, 1000 transactions/sec) & inspect Iceberg
python scripts/run_load_tests.py --rates 200,500,1000 --events-per-run 3000
python scripts/verify_iceberg.py --limit 10
```

```
# 5. Stop Docker containers when finished
docker compose down -v --remove-orphans
```